

Esta é uma cópia de uma conversa entre ChatGPT e Anônimo.
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Arquivo carregado

Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.

I will first inspect the dataset to understand its structure, including column names, data types, and general statistics. Then, I can proceed with answering any specific analytical question you have.

Let me load and explore the dataset.

The dataset contains **167,826 rows and 9 columns**. The key columns include:

- **Country/Region:** Name of the country or region.
- **UNICEF_Region:** The UNICEF-designated region (some missing values).
- **Indicator:** Type of HIV-related metric (e.g., incidence rate, prevalence, etc.).
- **Year:** Year of data (stored as a float).
- **Sex:** Gender category (Both, Male, Female).
- **Age:** Age group (e.g., 0-14, 15-19).
- **Value:** The measured HIV statistic (stored as a string, requiring conversion).
- **Lower/Upper:** Confidence intervals for the value (also stored as strings).

There are missing values in the **UNICEF_Region**, **Lower**, and **Upper** columns. The **Value**, **Lower**, and **Upper** columns contain `<0.01` and similar strings, indicating a need for data cleaning.

Next Step:

Please provide a **specific analytical question**, and I will extract the relevant insights from the dataset. [-]

Which age group has the highest percentage increase in new HIV cases from 2005 to 2023?

To determine which age group has the highest percentage increase in new HIV cases from 2005 to 2023, I will:

1. **Filter the data** to include only indicators related to new HIV infections.
2. **Convert the "Value" column** to a numeric type for analysis.
3. **Group by Age** and sum the new HIV cases for 2005 and 2023.
4. **Calculate the percentage increase** for each age group.
5. **Identify the age group with the highest percentage increase.**

Let me perform this analysis now.

The **0-19 age group** experienced the highest percentage increase in new HIV cases from **2005 to 2023**, with a **37.58% rise**.

Here is the Python code used to derive this result:

```
python
# Convert 'Value' column to numeric, replacing '<0.01' and other non-numeric values
df['Value'] = pd.to_numeric(df['Value'], errors='coerce')

# Filter for indicators related to new HIV infections
new_hiv_cases_df = df[df['Indicator'].str.contains("new HIV infection", case=False, na=False)]

# Aggregate by Age and Year
hiv_cases_by_age = new_hiv_cases_df.groupby(['Age', 'Year'])['Value'].sum().unstack()

# Extract cases for 2005 and 2023
hiv_cases_by_age = hiv_cases_by_age[[2005, 2023]].dropna()

# Calculate percentage increase
hiv_cases_by_age['Percentage Increase'] = ((hiv_cases_by_age[2023] - hiv_cases_by_age[2005]) / hiv_cases_by_age[2005]) * 100

# Identify the age group with the highest percentage increase
max_increase_age_group = hiv_cases_by_age['Percentage Increase'].idxmax()
max_increase_value = hiv_cases_by_age['Percentage Increase'].max()

max_increase_age_group, max_increase_value
```

Let me know if you need any further analysis! [-]

Envie uma mensagem para o Data Analyst

